THE UNIVERSITY OF MELBOURNE

Student number

Semester 2 Assessment, 2021

School of Mathematics and Statistics

MAST20030 Differential Equations

Reading time: 30 minutes — Writing time: 3 hours — Upload time: 30 minutes

This exam consists of 3 pages (including this page) with 7 questions and 100 total marks

Permitted Materials

- This exam and/or an offline electronic PDF reader and blank loose-leaf paper.
- One double sided A4 page of notes (handwritten only).
- No calculators are permitted. No headphones or earphones are permitted.

Instructions to Students

- Wave your hand right in front of your webcam if you wish to communicate with the supervisor at any time (before, during or after the exam).
- You must not be out of webcam view at any time without supervisor permission.
- You must not write your answers on an iPad or other electronic device.
- Off-line PDF readers (i) must have the screen visible in Zoom; (ii) must only be used to read exam questions (do not access other software or files); (iii) must be set in flight mode or have both internet and Bluetooth disabled as soon as the exam paper is downloaded.

Writing

- Write your answers on A4 paper. Page 1 should only have your student number, the subject code and the subject name. Write on one side of each sheet only. Each question should be on a new page. The question number must be written at the top of each page.

Scanning and Submitting

- **You must not leave Zoom supervision to scan your exam.** Put the pages in question order and all the same way up. Use a scanning app to scan all pages to PDF. Scan directly from above. Crop pages to A4.
- Submit your scanned exam as a single PDF file and carefully review the submission in Gradescope. Scan again and resubmit if necessary. Do not leave Zoom supervision until you have confirmed orally with the supervisor that you have received the Gradescope confirmation email.
- **You must not submit or resubmit after having left Zoom supervision.**

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Question 1 (12 marks)

(a) Consider the following second order ODE:

$$y''(t) + f_1(t)y'(t) + f_0(t)y(t) = 0. \tag{1}$$

Assume that you have found two linearly independent solutions $y_1(t), y_2(t)$ of (1). Show that the wronskian $\mathrm{wr}[y_1, y_2](t)$ satisfies a first order ODE and find the general solution of this ODE. *Explicit calculation required. Do not quote results from the lectures!*

(b) Now consider

$$ty''(t) - (4t+1)y'(t) + (4t+2)y(t) = 0. \tag{2}$$

(i) Find a solution $y_1(t)$ of (2) by trialling $y_1(t) = e^{\lambda t}$.

(ii) Use the wronskian to find the general solution of (2).

Question 2 (10 marks)

Use the matrix exponential to solve the following IVP:

$$\mathbf{y}\,'(t) = \frac{1}{4}\begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}\mathbf{y}\,(t), \qquad \mathbf{y}\,(1) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Question 3 (18 marks)

Consider the following second oder ODE:

$$(1-t)y''(t) + ty'(t) - y(t) = 0. \tag{3}$$

(a) Trial a power series $y(t) = \sum_{j=0}^{\infty} c_j t^j$ and derive a recursion for the series coefficients c_j.

(b) Compute the coefficients up to and including c_5 and deduce the solution of the recursion. You should find two linearly independent solutions $y_1(t)$, $y_2(t)$.

(c) Are the solutions $y_1(t), y_2(t)$ genuine? Justify your answer.

(d) Find the solution of the associated IVP with initial conditions $y(0) = y'(0) = 1$. Which function does this solution converge to? *You may use results from the lectures.*

(e) Now add an inhomogeneity to (3) so that the ODE becomes $(1-t)y''(t)+ty'(t)-y(t) = 5$. What is the *general* solution of this modified ODE?

Question 4 (18 marks)

(a) Find the following (inverse) Laplace transforms:

(i) $(\mathscr{L}(e^{3t}\cos^2 t - e^{3t}\sin^2 t))(s)$,

(ii) $(\mathscr{L}^{-1}\frac{e^{-2s}-s^2-s+4}{s^2+s-2})(t)$.

(b) Use Laplace transforms to solve the following IVPs:

(i)
$$y''(t) + 16y(t) = \delta_3^+(t) + 4H_2(t), \qquad y(0) = 6, \quad y'(0) = 0,$$

(ii)
$$y'(t) + 2y(t) = g(t), \qquad y(0) = 1,$$

First, find the solution for general $g(t)$ and then consider the two cases $g(t) = 1$ and $g(t) = e^{-2t}$.

Question 5 (10 marks)

(a) Determine the Fourier series of the function

$$f(x) = x - 1 \qquad 0 \le x \le 1,$$

extended to an even function on $-1 \le x \le 1$ and then periodically to $\mathbb{R}$.

(b) What does this Fourier series converge to? Is the convergence uniform? Explain your answer by quoting appropriate results from the lectures.

(c) Use your result to compute

$$\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2}.$$

Calculation required! Quoting results from lectures/tutorials is not enough.

Question 6 (22 marks)

Consider the inhomogeneous one-dimensional heat equation for $x \in [0,1]$, $t \ge 0$:

$$\frac{\partial u}{\partial t} = 3\frac{\partial^2 u}{\partial x^2} - 18x, \tag{4}$$

with inhomogeneous boundary conditions

$$\partial_x u(0,t) = -1, \qquad u(1,t) = -1, \qquad t > 0,$$

and the initial condition

$$u(x,0) = x^3 - 2x, \qquad 0 < x < 1.$$

(a) Trial a polynomial $u_{poly}(x) = \sum_{n=0}^{3} a_n x^n$ and determine $a_0, \ldots, a_3$ by requiring that $u_{poly}(x)$ satisfies the the inhomogeneous differential equation (4) **and** the boundary conditions.

(b) Now set $u(x,t) = u_c(x,t) + u_{poly}(x)$. Insert into (4) and confirm that the IBVP for $u_c(x,t)$ is the homogeneous heat equation with homogeneous boundary conditions and an initial condition of the form $u_c(x,0) = ax + b$, where the constants a and b are fixed by the result of (a).

(c) Solve the IBVP for $u_c(x,t)$ and thereby determine $u(x,t)$. *Note that you can answer this question without having done (a) and (b) by taking the statements of (b) and leaving the constants a, b and $u_{poly}(x)$ unfixed.*

(d) Is the solution genuine for $0 \le x \le 1$ and $t > 0$? Justify your answer.

(e) Does the IBVP for $u(x,t)$ have a steady state solution? If yes, identify it.

Question 7 (10 marks)

(a) Compute the Fourier transform of

$$f(x) = 2H_{-2}(x) - H_0(x) - H_1(x).$$

(b) Prove

$$(\mathscr{F}(f * g))(p) = (\mathscr{F} f)(p)(\mathscr{F} g)(p).$$

End of Exam — Total Available Marks = 100